Lecture 21 — Systems and $Ax = b$

Mon Oct 12

Last updated: October 8, 2026.

Boyd video (optional): Lecture 25 — linear equations · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 8.3. Related objectives: Lectures. Schedule. Connects to Ax as a linear combination of columns (Lecture 19).

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Translate equations or contextual constraints into $Ax = b$.
2. Interpret $Ax = b$ as asking whether b belongs to the span of the columns of A.
3. Classify systems as square, underdetermined, or overdetermined.
4. Anticipate circumstances producing zero, one, or many solutions.

Warm-up

Rewrite the system

$$x_1 - 2x_2 = 3$$
$$2x_1 + x_2 = 1$$

as $Ax = b$. What are A, x, and b? Is b in the span of the columns of A? (You may decide by solving or by a short span argument.)

Fill in

A linear system with m equations in n unknowns can be written $Ax = b$, where A is _____, x is an _____-vector, and b is an _____-vector.

Column view. Solving $Ax = b$ asks: is b a _____ of the columns of A? Equivalently: does b lie in span{columns of A}?

Shape vocabulary (compare m equations to n unknowns):

Name	Relation	Typical intuition
Square	$m =$ ____	Often one solution, but not always
Overdetermined	$m >$ ____	Often _____ (too many constraints)

Underdetermined	$m <$ ____	Often _____ (free variables)

These are **tendencies**, not guarantees: a square system can be inconsistent or have infinitely many solutions; an overdetermined system can still be consistent.

Possible outcomes for $Ax = b$:

1. **No solution** — b is _____ the column space.
2. **Unique solution** — b is in the column space in exactly _____ way.
3. **Infinitely many solutions** — b is in the column space in _____ ways (free parameters).

Worked in class

W1. Matrix form. Three unknowns x_1, x_2, x_3:

$$\begin{aligned} x_1 + x_2 &= 4 \\ x_2 - x_3 &= 1 \\ 2x_1 + x_3 &= 3. \end{aligned}$$

Write A, x, and b. Square / over / under?

W2. Span interpretation. Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

(a) Are the columns of A independent? Describe span{columns}. (b) Does $Ax = b$ have a solution? Infinitely many? (c) Does $Ax = c$ have a solution?

W3. Context $\to Ax = b$. A café sells tea (\$2) and coffee (\$3). On Monday they sold x_1 teas and x_2 coffees for \$26 total revenue, and sold 10 drinks in all. Write the two equations and the matrix form. Square?

W4. Anticipating outcomes. Without fully solving, say what you expect (and why) for:

(a) 2 equations, 2 unknowns, with clearly parallel incompatible lines.

(b) 1 equation, 3 unknowns: $x_1 + x_2 + x_3 = 5$.

(c) 3 equations, 2 unknowns that happen to be consistent (e.g. the third is the sum of the first two).

Your turn

Y1. Write as $Ax = b$:

$$3x_1 - x_2 = 0$$
$$x_1 + 4x_2 = 7.$$

Classify as square / over / under.

Y2. Let A have columns $a_1 = (1, 0, 1)$ and $a_2 = (0, 1, 1)$. For $b = (1, 1, 2)$, explain in one sentence why $Ax = b$ has a

solution, and give one solution x.

Y3. True or false, with a one-line reason: every square system $Ax = b$ has exactly one solution.

Y4. A sensor gives 4 linear measurements for 2 unknown parameters. Is the system typically overdetermined? Does that guarantee inconsistency?

Answers (Your turn)

Y1.

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 7 \end{bmatrix}.$$

Square ($m = n = 2$).

Y2. $b = 1 \cdot a_1 + 1 \cdot a_2$, so b is in the column span; one solution is $x = (1, 1)$.

Y3. **False.** Square systems can be inconsistent (parallel lines) or have infinitely many solutions (same line twice).

Y4. Typically overdetermined ($m = 4 > n = 2$). No — consistency depends on whether b lands in a 2-dimensional column space; it might or might not.